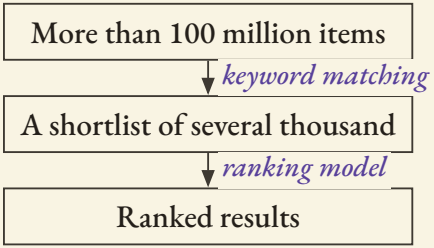
Etsy: a shortlist out of 100 million items

Etsy is an online marketplace for handmade, vintage and craft items.

For each search it first shortlists several thousand items, then ranks the shortlist.

The shortlist used to come from keyword matching only, which fails when buyers and sellers use different words.



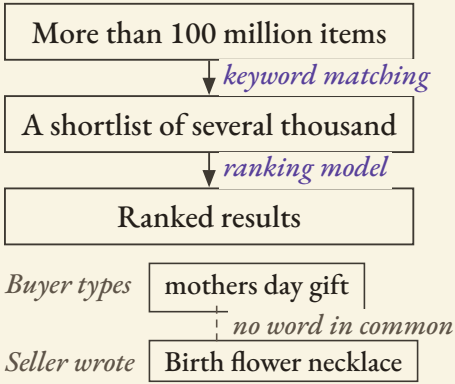
Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024. The necklace listing is invented.

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Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024. The necklace listing is invented.

The training data: a month of search logs

One row per search that ended in a cart add or a purchase.

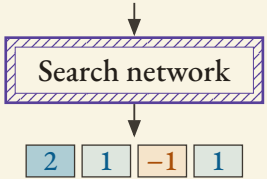
1 month: more than 150 million rows, from up to 30 million users.

Search	Who searched	Item	Action
mothers day gift	Zurich, searched necklaces	Flower necklace	purchase
bridge photos	New York, clicked print shops	Bridge print	cart add
wedding guest book	Porto, bought wedding decor	Guest book	purchase
wool socks	Oslo, searched knitting	Knitted socks	cart add

The rows are invented; the counts are Etsy's. Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024.

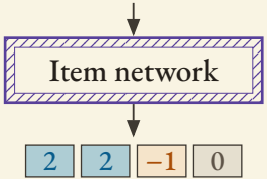
Every search and every item becomes a list of numbers

The search words, the location, recent searches and shops clicked



mothers day gift, a user in Zurich

The item's title, tags and category, and the searches that led to it before

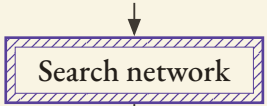


Flower necklace

Invented numbers. Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024.

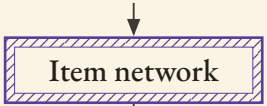
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Flower necklace

Training moves the numbers until the inner product is high for the item bought after the search, and low for random items. Here it is 7.

Invented numbers. Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024.

The new search: sort the items by inner product

mothers day gift 2 1 −1 1

Invented numbers.

The new search: sort the items by inner product

mothers day gift	2	1	−1	1

Gift wrap	−1	0	2	1
Flower necklace	2	2	−1	0
Bridge print	0	−1	1	−2
Photo mug	1	1	0	1
Recipe board	1	2	0	1

Invented numbers.

The new search: sort the items by inner product

mothers day gift	2	1	−1	1	*inner product*
	×	×	×	×	
Flower necklace	2	2	−1	0	
	4	+ 2	+ 1	+ 0	= 7
Gift wrap	−1	0	2	1	
Flower necklace	2	2	−1	0	
Bridge print	0	−1	1	−2	
Photo mug	1	1	0	1	
Recipe board	1	2	0	1	

Invented numbers.

The new search: sort the items by inner product

					inner product
mothers day gift	2	1	−1	1	
	×	×	×	×	
Flower necklace	2	2	−1	0	
	4	+ 2	+ 1	+ 0	= 7
Gift wrap	−1	0	2	1	−3
Flower necklace	2	2	−1	0	7
Bridge print	0	−1	1	−2	−4
Photo mug	1	1	0	1	4
Recipe board	1	2	0	1	5

Invented numbers.

The new search: sort the items by inner product

					inner product
mothers day gift	2	1	−1	1	
	×	×	×	×	
Flower necklace	2	2	−1	0	
	4	+ 2	+ 1	+ 0	= 7
Gift wrap	−1	0	2	1	−3
Flower necklace	2	2	−1	0	7
Bridge print	0	−1	1	−2	−4
Photo mug	1	1	0	1	4
Recipe board	1	2	0	1	5

Invented numbers.

Sorted by inner product; the top three join the shortlist

Flower necklace	7
Recipe board	5
Photo mug	4
Gift wrap	−3
Bridge print	−4

What it cost and what it paid

Cost

72 hours *of training, on 4 GPUs*

1 month *of cart adds and purchases*

Value: A/B tests, relative change

+2.63% *conversion rate of the whole site*

+5.58% *purchase rate of searches*

Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024.

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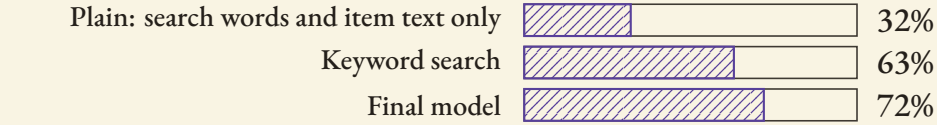
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Popular searches: the share of the products bought that a version had in its top 100



Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024.